

Irrationality after One Stabilization of Universally CH_0 -Trivial Cubic Threefolds

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Abstract

Does universal CH_0 -triviality force rationality after stabilization? We prove that $X \times \mathbf{P}^1$ is irrational for every smooth cubic threefold X . Yang–Yu–Zhu construct a two-dimensional family of smooth cubic threefolds with unirational parametrizations of coprime degrees, hence universally CH_0 -trivial; with our theorem that family already separates the two properties after one stabilization. The simplest instance is the Fermat fourfold: $\{x_1^3 + \cdots + x_5^3 = 0\} \times \mathbf{P}^1$ is universally CH_0 -trivial and irrational. We exhibit a second, non-isotrivial pencil of universally CH_0 -trivial cubic threefolds X_b with $X_b \times \mathbf{P}^1$ irrational, on which the vanishing of the diagonal-decomposition obstruction is proved by an integral lattice saturation on the intermediate Jacobian rather than by a unirational parametrization or an isogeny to a curve Jacobian.

The cycle argument proves an all-degree integral saturation theorem for marked finite-etale graph quotients of elliptic powers: divided powers of the marked divisor lattice are ordinary divisor products. The six-axis intermediate Jacobians in the nonstandard A_5 -invariant cubic pencil satisfy its hypotheses. Voisin’s theorem gives universal CH_0 -triviality on every smooth fibre.

For irrationality, we count primitive-sixth eigenvalues of framed formal monodromy on the numerical small even quantum connection. Divisor tags make Iritani’s blowup formula compatible with a possibly noninjective center specialization. The count is a birational invariant through dimension four and equals two for a cubic. Thus the cycle-theoretic and quantum detectors diverge on the same smooth fourfolds.

Keywords. cubic threefold; universal CH_0 -triviality; irrationality; intermediate Jacobian; quantum connection; weak factorization.

1 Introduction

A smooth projective variety is universally CH_0 -trivial if the degree map

$$\deg : CH_0(Y_L) \longrightarrow \mathbf{Z}$$

is an isomorphism for every field extension $L/\mathbf{C}$. This condition is a central consequence of stable rationality, and its failure is a standard stable-rationality obstruction. Its validity, however, need not provide a rational parametrization. We ask how far that gap can persist after one explicit stabilization.

The irrationality half of that question has a uniform answer. The small even quantum connection of every smooth cubic threefold has primitive-sixth framed formal monodromy, and Iritani’s quantum blowup formula shows that its multiplicity survives one stabilization.

Theorem 1.1 (One-step irrationality). *For every smooth complex cubic threefold X , the fourfold $X \times \mathbf{P}^1$ is irrational.*

The conclusion depends only on the birational class: if Y is birational to a smooth cubic threefold then $Y \times \mathbf{P}^1$ is birational to $X \times \mathbf{P}^1$, so Theorem 1.1 applies to Y as well. This covers the Fano threefolds of degree fourteen and Picard number one, and the blow-ups of X in a line or in a plane cubic curve, listed in [22, Remark 3.7]. For genus eight the invariant itself is computed: Corollary 4.13 gives $\nu_6 = 2$ there, without using the classical birationality with a cubic.

Universally CH_0 -trivial cubic threefolds are known in abundance, and Theorem 1.1 turns every such locus into a locus where the two properties separate. Voisin proves that the moduli space of smooth cubic threefolds, which has dimension ten, contains a nonempty countable union of subvarieties of codimension at most three parametrizing threefolds with universally trivial CH_0 [20, Theorem 4.5].

Corollary 1.2 (Separation in codimension three). *There is a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds such that, for every member X , both X and $X \times \mathbf{P}^1$ are universally CH_0 -trivial while $X \times \mathbf{P}^1$ is irrational.*

One member is completely explicit. Colliot-Thélène proves that a smooth cubic hypersurface whose equation is a sum of forms in separated groups of at most three variables is universally CH_0 -trivial [6], which applies to the Fermat equation directly.

Corollary 1.3 (Fermat). *The fourfold $\{x_1^3 + x_2^3 + x_3^3 + x_4^3 + x_5^3 = 0\} \times \mathbf{P}^1$ is universally CH_0 -trivial and irrational.*

Yang–Yu–Zhu construct a two-dimensional family of smooth cubic threefolds admitting unirational parametrizations of degrees two and three [22, Theorems 1.1 and 3.3]. Unirational parametrizations of coprime degrees give a decomposition of the diagonal, so every member of that family is universally CH_0 -trivial, and the same parametrizations persist on $X \times \mathbf{P}^m$ for every m [22, Corollary 3.5].

Corollary 1.4 (Coprime-degree cubics). *Let X be one of the cubic threefolds of [22, Theorem 3.3]. Then $X \times \mathbf{P}^1$ admits unirational parametrizations of coprime degrees and is universally CH_0 -trivial, and it is irrational.*

Yang–Yu–Zhu ask whether a variety with unirational parametrizations of coprime degrees can fail to be stably rational [22, Question 1.4]. Corollary 1.4 verifies the first non-trivial stabilization of that question for their examples. It does not settle it: rationality of $X \times \mathbf{P}^m$ for some $m \geq 2$ is not excluded, for the reason recorded after Theorem 1.5.

Our second result concerns a family whose universal CH_0 -triviality is proved by neither route. Here X is a smooth cubic threefold in the nonstandard A_5 -invariant pencil, and the primitive minimal class of its intermediate Jacobian is algebraic by an integral saturation theorem for the six-axis lattice.

Theorem 1.5 (Separation on the A_5 -pencil). *Let W_5 be the five-dimensional irreducible A_5 -module, and let B° be the smooth locus inside*

$$\mathbf{P}((\mathrm{Sym}^3 W_5^*)^{A_5}).$$

For the resulting family $\mathcal{X} \rightarrow B^\circ$ and every $b \in B^\circ(\mathbf{C})$,

$$\mathcal{X}_b \text{ is universally } CH_0\text{-trivial,} \quad \mathcal{X}_b \times \mathbf{P}^1 \text{ is universally } CH_0\text{-trivial and irrational.}$$

The family is non-isotrivial in coarse moduli.

The invariant cubic space in Theorem 1.5 is two-dimensional. Indeed, the character of W_5 is $(5, 1, -1, 0, 0)$ on the classes $1, 2A, 3A, 5A, 5B$. The symmetric-cube character formula gives

$$\dim(\mathrm{Sym}^3 W_5^*)^{A_5} = 2.$$

Thus its projectivization is a pencil with connected smooth locus B° . Hartlieb denotes it by M_{H_1} ; it is not the component associated with $\mathbf{1} \oplus W_4$ [9, Lemma 5.5]. Since the family is non-isotrivial, its image in coarse moduli is a curve, one dimension below the locus of [22].

Remark 1.6. The pencil contains the Fermat cubic threefold; it is one of the two members that also admit the permutation action of A_5 [9, Lemma 5.5]. The mechanism is that W_5 is induced from a nontrivial cubic character χ of a point stabilizer $A_4 \subset A_5$: Frobenius reciprocity gives

$$\langle \mathrm{Ind}_{A_4}^{A_5} \chi, W_5 \rangle = \langle \chi, \mathrm{Res}_{A_4} W_5 \rangle = 1,$$

and both sides have dimension five, so $\mathrm{Ind}_{A_4}^{A_5} \chi \cong W_5$. Induction from a character realizes W_5 by monomial matrices whose nonzero entries are cube roots of unity, and such a matrix fixes $x_1^3 + \cdots + x_5^3$. Corollary 1.3 is therefore the member of Theorem 1.5 at which universal CH_0 -triviality is also classical, and Yang–Yu–Zhu give that member an explicit degree-three unirational parametrization [22, Remark 3.6].

We do not claim that X is stably irrational. From the second stabilization onward, weak-factorization centers can themselves have cubic dimension, and the low-dimensional exclusion used here no longer applies.

How the two proofs meet

The cycle argument starts from six elliptic quotients of the Fano surface. Their Rosati Gram matrix is $6I - J$, and the geometric kernel lies in the exotic two-point complement of $\mathbf{P}^1(\mathbf{F}_2) \subset \mathbf{P}^1(\mathbf{F}_4)$. Its local graph slopes are finite etale at two and three. The graph equations then make the descended divisor lattices rank-one generated. Since rank-one coefficient divisors have square zero, $\Theta^4/4!$ is an ordinary integral divisor product.

The quantum argument counts primitive sixth-root eigenvalues of framed formal monodromy on the numerical small even quantum connection. Iritani’s operation formulas make this count additive through weak factorization; divisor tags repair a potentially noninjective center specialization. Starting from Cai’s displayed small-even matrix, the internal calculation gives the nonzero count, while every center of dimension at most two contributes zero. The special A_5 -geometry enters only in the cycle argument; the quantum theorem applies to every smooth cubic threefold.

Relation with earlier rationality results

Clemens and Griffiths proved that every smooth complex cubic threefold is irrational by means of its intermediate Jacobian [5]. That obstruction does not by itself control a product with projective space, and stable rationality of smooth cubic threefolds remains open. Cai recently recovered irrationality from the primitive-sixth-root formal monodromy of the cubic quantum connection [4]. Iritani and Iritani–Koto prove the quantum D -module decompositions for blowups and projective bundles [11, 13]; Katzarkov–Kontsevich–Pantev–Yu identify the nef-canonical normal form used for low-dimensional centers [14]. Theorem 1.1 combines these inputs at the next dimension.

On the cycle side, Voisin reduces universal CH_0 -triviality of a cubic threefold to algebraicity of the primitive minimal class of its intermediate Jacobian [20]. Beckmann–de

Gaay Fortman place such minimal classes in a general integral-Fourier framework [3]. For the very general cubic threefold the minimal class is not algebraic: every curve class on the intermediate Jacobian is an even multiple of it, so a very general cubic threefold admits no decomposition of the diagonal [7, Theorem 1.3 and Corollary 1.4]. What Voisin's criterion leaves is the countable union of moduli subvarieties on which the class is algebraic: the codimension-three components used in Corollary 1.2 [20, Theorem 4.5 and Lemma 4.6], together with the almost-diagonal cubics of [6] and the coprime-degree family of [22]. Theorem 3.5 does not enlarge that union. It proves algebraicity uniformly along one explicit pencil, from the six-axis lattice rather than from an isogeny to a curve Jacobian, from separated variables, or from a unirational parametrization. The six elliptic quotients supplied by Roulleau and the special period locus isolated by Hartlieb make the class explicit: it belongs to the ordinary integral divisor-product lattice for every smooth member of the nonstandard A_5 -pencil.

Proof map

Section 2 computes the six-axis polarization and identifies the exotic cubic gluing. Section 3 proves the local divisor-product theorem and applies it to the primitive minimal class. Section 4 proves Theorem 1.1 from formal monodromy and weak factorization. The final section combines the two detectors and records the exact scope boundary.

2 The six-axis polarization and the exotic packet

Let Ω be the six conjugate D_5 -subgroups of A_5 , equivalently the six Sylow-5 subgroups with their normalizers, and put

$$H_p = \text{Aug}(\mathbf{F}_p^\Omega) / \langle \mathbf{1} \rangle \quad (p = 2, 3).$$

The modules H_2, H_3 are simple, with $\text{End}_{A_5}(H_2) = \mathbf{F}_4$ and $\text{End}_{A_5}(H_3) = \mathbf{F}_3$ [10, the A_5 matrix-representation rows in characteristic two, GF(4)-dimension two (IDs a,b) and GF(2)-dimension four (IDs a,b), and in characteristic three, GF(3)-dimension four].

The six axes arise from the Fano surface. Let S be the Fano surface of a smooth member of the A_5 -pencil and $(J, \Theta) = \text{Alb}(S)$ its intermediate Jacobian. Each of the six D_5 -subgroups gives an elliptic quotient $q_H : J \rightarrow E_H$ and its Rosati dual $i_H : E_H \rightarrow J$.

Generic non-CM axis. Hartlieb's elliptic isogeny factor, and hence every generic six-axis elliptic quotient, is non-CM. Indeed, a CM endomorphism has a fixed imaginary-quadratic order, so the generic j -invariant is a root of its Hilbert class polynomial and is constant. After a finite base change the elliptic factor would then be isotrivial. Hartlieb's isogeny $J_b \sim E_b^5$ would put the whole intermediate-Jacobian family in the countable isogeny class of one fixed abelian variety. The image of an algebraic curve in a moduli space cannot be both positive-dimensional and countable. This contradicts the fact that the closure of the intermediate-Jacobian image is a one-dimensional special subvariety [9, Proposition 5.7 and Remark 5.8]. Since the generic intermediate Jacobian is isotypic to E_b^5 , each nonzero elliptic quotient is isogenous to E_b , and so is non-CM as well.

Proposition 2.1 (Six-axis polarization). *After coherent identification of the elliptic factors, the Rosati Gram matrix of the six maps is*

$$G_6 = 6I_6 - J_6.$$

Any five axes therefore give an isogeny $E^5 \rightarrow J$ whose pulled-back polarization has Smith type $(1, 6, 6, 6, 6)$.

Proof. We recall the numerical passage from Roulleau's genus-two configuration to the six elliptic fibres. Let $D_g \subset S$ be the curve indexed by an involution $g \in A_5$, put $D_{\text{tot}} = \sum_g D_g$, and, for a dihedral subgroup H of order ten, put

$$F_H = \sum_{g \in H, g^2=1} D_g.$$

Roulleau proves that F_H is a fibre of $S \rightarrow E_H$ [18, Theorem 11(D)], so $F_H^2 = 0$. His intersection table has $D_g^2 = -4$, and distinct curves meet with multiplicity zero, two, or one when the product of their labels has order two, three, or five, respectively; Lemma 14 proves that these values are constant in the corresponding families [18, Lemma 14, Corollary 16, and (3.1)]. Among the 105 unordered pairs of involutions, the product orders two, three, and five occur 15, 30, 60 times. Hence

$$D_{\text{tot}}^2 = 15(-4) + 2(30 \cdot 2 + 60 \cdot 1) = 180.$$

Every involution belongs to two of the six dihedral subgroups, and therefore $\sum_H F_H = 2D_{\text{tot}}$. Two-transitivity on the six subgroups now gives

$$30(F_H \cdot F_{H'}) = (2D_{\text{tot}})^2 = 720, \quad F_H \cdot F_{H'} = 24 \quad (H \neq H').$$

By the preceding generic non-CM argument, the elliptic factor has only integer endomorphisms. After coherent identification of the generic elliptic quotients, two-transitivity makes the Rosati matrix have diagonal d and off-diagonal m . Put

$$D_H = q_H^*[0] \in \text{NS}(J), \quad N_H = i_H q_H \in \text{End}(J).$$

Then $D_H|_S = F_H$, and N_H is the Rosati endomorphism attached to D_H . Coherent identification gives $q_H i_H = [d]$ and $q_H i_{H'} = [m]$ for $H \neq H'$. The invariant sum of the six axis maps vanishes because the five-dimensional constituent has no trivial summand; composing with q_H gives $d + 5m = 0$. The minimal-class identity $[S] = \Theta^3/3!$ [5], together with the polarized Riemann–Roch trace formula

$$\frac{\alpha\beta\Theta^{g-2}}{(g-2)!} = \text{tr}(N_\alpha) \text{tr}(N_\beta) - \text{tr}(N_\alpha N_\beta),$$

obtained by polarizing the reduced-norm Riemann–Roch polynomial [8, Theorem 4.1], gives the following identities. Here tr is the degree-five trace in the coefficient representation—the trace entering that reduced norm—rather than the degree-ten trace on rational first homology:

$$\text{tr}(N_H) = d, \quad \text{tr}(N_H N_{H'}) = m^2, \quad F_H F_{H'} = d^2 - m^2.$$

Thus $d^2 - m^2 = 24$. Together with $d + 5m = 0$ and $d > 0$, this forces $(d, m) = (5, -1)$, hence $G_6 = 6I - J$. A five-by-five minor has Smith form $(1, 6, 6, 6, 6)$. $\square$

Let B° be the connected smooth locus of the nonstandard A_5 -pencil, and let $\mathcal{J} \rightarrow B^\circ$ be its relative intermediate Jacobian.

Lemma 2.2 (Relative six-axis source). *There are an elliptic scheme $\mathcal{E} \rightarrow B^\circ$ and a relative isogeny*

$$f : \mathcal{A} := \mathcal{E} \otimes_{\mathbf{Z}} (\mathbf{Z}^\Omega / \mathbf{Z}\mathbf{1}) \longrightarrow \mathcal{J}$$

which is A_5 -equivariant. The quotient lattice carries the form induced by $6I_6 - J_6$; both the source and this form carry their natural S_6 -action. Moreover,

$$f^* \lambda_\Theta = \lambda_{\mathcal{A}}. \tag{2.1}$$

After omitting one axis, $\mathcal{A} \simeq \mathcal{E}^5$ and this form has matrix $6I_5 - J_5$. For $p = 2, 3$, put $\mathcal{V}_p = \mathcal{E}[p]$. The p -primary kernel of the source polarization has a natural A_5 -equivariant symplectic description

$$\mathcal{D}_p \simeq H_p \otimes_{\mathbf{F}_p} \mathcal{V}_p \quad (p = 2, 3). \quad (2.2)$$

The p -primary part $\mathcal{K}_p$ of $\ker f$ is an A_5 -stable relative maximal isotropic subgroup of $\mathcal{D}_p$.

Proof. For $H \in \Omega$, the integral norm $n_H = \sum_{h \in H} h$ is a relative endomorphism of $\mathcal{J}$. Its action on rational first homology is ten times the projector onto the H -fixed summand. Fibrewise, Hartlieb's character calculation and isogeny decomposition give $H^1(J, \mathbf{Q}) \simeq W_5 \otimes H^1(E, \mathbf{Q})$, and hence the self-product of an elliptic curve through the five-dimensional module W_5 [9, Lemma 5.5 and Remark 5.8]. Since $\dim W_5^H = 1$, the fixed summand has rank two. More concretely, $n_H^2 = 10n_H$. The scheme-theoretic image of a homomorphism of abelian schemes in characteristic zero is an abelian subscheme, and its quotient factorization commutes with base change [2, Theorem A]. Thus the scheme-theoretic image $\mathcal{E}_H \subset \mathcal{J}$ is an abelian scheme of relative dimension one. It is therefore an elliptic scheme. Since $hn_H = n_H$ for every $h \in H$, the subgroup H acts trivially on $\mathcal{E}_H$. Moreover $N_{A_5}(H) = H$, so transport by an element carrying H to H' is independent of the chosen transporter. The six elliptic schemes are therefore coherently isomorphic; fix one of them as $\mathcal{E}$.

Write $i_H : \mathcal{E} \rightarrow \mathcal{J}$ for the transported primitive inclusions. On the geometric generic fibre, Hartlieb's isogeny decomposition identifies $\mathrm{Hom}^0(E, J)$ with the irreducible module W_5 . The coherently transported maps $(i_H)_{H \in \Omega}$ form one A_5 -orbit, so their sum is invariant. Since $W_5^{A_5} = 0$, that sum is zero. The connected smooth curve B° is integral, $\mathcal{E}$ is integral, and $\mathcal{J}$ is separated. Equality on the generic fibre therefore gives $\sum_H i_H = 0$ over B° . Hence $(i_H)_{H \in \Omega}$ descends from $\mathcal{E}^\Omega$ to the displayed map $f : \mathcal{A} \rightarrow \mathcal{J}$, and the construction is A_5 -equivariant.

Rouleau constructs the D_5 -fibration from an Albanese quotient $q_H : J \rightarrow E_H$ by a connected four-dimensional abelian subvariety [18, Lemma 17 and the following proof of Theorem 11(D)]. Thus $F_H = q_H^*[0]|_S$. The connected kernel makes the Rosati dual of q_H a primitive inclusion, and its image is the unique H -fixed elliptic subvariety. It therefore agrees, up to an automorphism of the target elliptic curve, with the norm image $\mathcal{E}_H \hookrightarrow \mathcal{J}$. This also identifies the norm-image axis, up to isogeny, with Hartlieb's generic elliptic isogeny factor. Consequently the fibre classes in Proposition 2.1 compute these relative homomorphisms, rather than merely isogenous quotients. On the generic non-CM fibre they give diagonal entry 5 and off-diagonal entry -1 . Applying the same generic-density and separatedness argument to these homomorphisms gives the identities over B° , hence (2.1). Choosing any five coordinate classes identifies the quotient lattice with $\mathbf{Z}^5$, where its matrix is $6I_5 - J_5$. Since this has Smith type $(1, 6, 6, 6, 6)$, the source polarization is nondegenerate, so f is a finite flat isogeny. From $\lambda_{\mathcal{A}} = f^\vee \lambda_\Theta f$ one gets

$$|\ker \lambda_{\mathcal{A}}| = 6^8, \quad |\ker f| = \deg f = 6^4.$$

For $x, y \in \ker f$, functoriality of the polarization commutator pairing gives $e_{\lambda_{\mathcal{A}}}(x, y) = e_{\lambda_\Theta}(fx, fy) = 1$. Thus $\ker f$ is isotropic and has square-root order in $\ker \lambda_{\mathcal{A}}$, hence is maximal isotropic. Its p -primary part has fibrewise order p^4 for $p = 2, 3$.

For $p = 2, 3$, the primary discriminant local system is

$$\mathcal{D}_p = \ker(6I_6 - J_6 : \mathcal{A}[p] \longrightarrow \mathcal{A}^\vee[p]).$$

Modulo p , the radical quotient of the induced coefficient form is canonically

$$\mathrm{Aug}(\mathbf{F}_p^\Omega)/\langle \mathbf{1} \rangle = H_p.$$

The form on this coefficient discriminant is the normalized form

$$b_p(\bar{x}, \bar{y}) = \frac{1}{p} x^\dagger (6I_6 - J_6) y \pmod{p} = \begin{cases} \bar{x} \cdot \bar{y}, & p = 2, \\ -\bar{x} \cdot \bar{y}, & p = 3. \end{cases} \quad (2.3)$$

For augmentation lifts, the product of the coordinate sums is divisible by p^2 , which proves the displayed formulas. They are unchanged by adding the constant vector or p times an integral vector. The dot product on the augmentation module has radical exactly the constant line, so b_p is nondegenerate on H_p ; b_2 is alternating and b_3 is symmetric. Tensoring with the Weil pairing on $\mathcal{V}_p$ gives (2.2) as a symplectic isomorphism. Since f is A_5 -equivariant, each $\mathcal{K}_p$ is A_5 -stable. The preceding order and isotropy calculation makes it a relative maximal isotropic subgroup of this rank-eight local system. $\square$

Proposition 2.3 (Principal gluing packet). *The A_5 -stable two-primary principal kernels of the six-axis source are parametrized by*

$$\mathbf{P}^1(\mathbf{F}_4) = \mathbf{P}^1(\mathbf{F}_2) \sqcup \{\omega, \omega^2\}.$$

The three $\mathbf{F}_2$ -rational kernels are S_6 -stable. Each of the other two has stabilizer exactly A_5 , and the outer normalizer exchanges them. The intermediate Jacobian of the nonstandard A_5 -cubic component uses the latter pair. The A_5 -stable three-primary kernels form $\mathbf{P}^1(\mathbf{F}_3)$; after choosing a complementary symplectic ruling, each is a scalar graph.

Proof. At two the coefficient discriminant is the simple four-dimensional $\mathbf{F}_2 A_5$ -module H_2 , whose commutant is $\mathbf{F}_4$. The full discriminant is the semisimple module $H_2 \otimes_{\mathbf{F}_2} \mathbf{F}_2^2 \simeq H_2^{\oplus 2}$. Every stable half is therefore one copy of H_2 , and Schur's lemma identifies such submodules with the one-dimensional $\mathbf{F}_4$ -subspaces of $\mathbf{F}_4^2$. This gives $\mathbf{P}^1(\mathbf{F}_4)$.

Here is the polarization check explicitly. Identify the coefficient heart with the natural two-dimensional $\mathbf{F}_4$ -module. The trace-determinant form is a nonzero invariant alternating form. Relative to it, the normalized form b_2 of (2.3) differs by a nonzero scalar in $\mathbf{F}_4$; rescaling an $\mathbf{F}_4$ -basis absorbs that scalar because squaring permutes $\mathbf{F}_4^\times$. We may therefore put

$$b(x, y) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2} \det(x, y).$$

On two multiplicity copies use

$$\Omega((x, y), (x', y')) = b(x, y') + b(y, x').$$

Every scalar $a \in \mathbf{F}_4$ is self-adjoint for b , since $b(ax, y) = b(x, ay)$. Thus the vertical line and all four graphs Γ_a are isotropic of half dimension. The five module-theoretic possibilities are exactly the five polarized gluings; no further quadratic condition removes the exotic pair.

The simplex group acts on both multiplicity copies of the coefficient heart over $\mathbf{F}_2$. Hence the three lines defined over $\mathbf{F}_2$ retain S_6 . The stabilizer of an exotic graph is equally intrinsic. A permutation that fixes Γ_ω commutes with ω , hence is $\mathbf{F}_4$ -linear on the two-dimensional $\mathbf{F}_4$ -space H_2 . If its determinant is $d \in \mathbf{F}_4^\times$, preservation of b says

$$\mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(dz) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(z) \quad (z \in \mathbf{F}_4).$$

The trace pairing is nondegenerate, so $d = 1$. It therefore lies in $\mathrm{SL}_2(4)$, of order 60. The given A_5 already has that order and lies in the stabilizer, proving equality. The odd coset in its order-120 normalizer is semilinear and applies field Frobenius, so it exchanges ω and ω^2 .

To pass from the generic fibre to the family, put

$$\mathcal{P} = \mathbf{P}_{\mathbf{F}_4}(\mathbf{F}_4 \otimes_{\mathbf{F}_2} \mathcal{V}_2).$$

By Lemma 2.2, this is the five-sheet local system of A_5 -stable principal halves. It has the degree-three sub-local system $\mathcal{P}_{\text{rat}} = \mathbf{P}_{\mathbf{F}_2}(\mathcal{V}_2)$ and the degree-two complement $\mathcal{P}_{\text{ex}}$. Here H_2 , together with its commutant $\mathbf{F}_4$, is constant, while monodromy acts only on $\mathcal{V}_2$. Its action is therefore $\mathbf{F}_4$ -linear, not semilinear, and factors through $\text{GL}_2(\mathbf{F}_2)$; in particular it preserves both subsets. The relative kernel $\mathcal{K}_2$ gives a section of $\mathcal{P}$.

At the geometric generic point this section cannot be rational. Otherwise the natural faithful S_6 -action on $\mathcal{A} = \mathcal{E} \otimes (\mathbf{Z}^\Omega / \mathbf{Z}\mathbf{1})$, which is the identity on $\mathcal{E}$ tensored with the faithful permutation action on the quotient lattice, would preserve the rational kernel: under (2.2) it acts on H_2 and trivially on $\mathcal{V}_2$, so every member of $\mathcal{P}_{\text{rat}}$ is S_6 -stable. At three, the analogous coefficient heart H_3 is simple and $\text{End}_{A_5}(H_3) = \mathbf{F}_3$. The semisimple module $H_3^{\oplus 2}$ has stable halves parametrized by $\mathbf{P}^1(\mathbf{F}_3)$. The tensor pairing is b_3 times the alternating multiplicity pairing, so the three affine scalar graphs and the vertical line are all maximal isotropic. The natural S_6 -action is the same on both multiplicity copies, so it fixes every one of these four lines. Since $|\ker f| = 6^4$, its two- and three-primary parts exhaust the kernel. The full kernel would therefore be S_6 -stable. Since this action preserves $6I_6 - J_6$, it would descend through the principally polarized quotient. It would act faithfully: if a source automorphism acts trivially on the quotient, the image of its difference from the identity is both connected and contained in the finite isogeny kernel, hence zero. Strong Torelli gives an exact sequence [9, Theorem 3.1]

$$1 \longrightarrow \{\pm 1\} \longrightarrow \text{Aut}(J, \Theta) \longrightarrow \text{Aut}(X) \longrightarrow 1.$$

The kernel of the resulting map from S_6 to $\text{Aut}(X)$ would be a normal subgroup of order at most two; it is trivial. Thus S_6 would act faithfully on a smooth cubic threefold. This contradicts the classification of faithful automorphism groups [9, Theorem 2.1]. Five of the six listed ambient groups have order smaller than $|S_6| = 720$. The remaining group is $(\mathbf{Z}/3\mathbf{Z})^4 \rtimes S_5$, of order 9720, which is not divisible by 720. None can contain S_6 . The section is therefore generically in $\mathcal{P}_{\text{ex}}$. Its inverse image is open and closed in the connected base B° , so every smooth fibre has exotic two-primary kernel. The same calculation at three gives the four lines parametrized by $\mathbf{P}^1(\mathbf{F}_3)$; choosing any complementary line as the vertical ruling makes the selected half an affine scalar graph. $\square$

For the cycle-theoretic argument, the unordered pair is enough. Both members act on the heart with the same squarefree minimal polynomial $t^2 + t + 1$. Theorem 3.5 uses only the conclusion that the geometric kernel is not one of the three $\mathbf{F}_2$ -rational gluings. At three it uses only scalarity of the possible A_5 -stable graphs, not the finer monodromy selection.

Hartlieb proves that the closure of its intermediate-Jacobian image is a one-dimensional special subvariety [9, Proposition 5.7]. Since the cubic period map is a locally closed embedding, the resulting smooth family is non-isotrivial. Only this landing and nonconstancy are used below; no identification of compactified parameter curves is needed for the separation theorem.

3 Integral divisor products and the primitive minimal class

Let (A, Θ) be a principally polarized abelian variety of dimension g . Put

$$P_A^k = \text{im}(\text{Sym}^k \text{NS}(A) \longrightarrow H^{2k}(A, \mathbf{Z})), \quad \gamma_A = \frac{\Theta^{g-1}}{(g-1)!}.$$

The class $D^k/k!$ of a divisor on an abelian variety is integral. The question here is whether these divided powers already belong to the smaller lattice P_A^k generated by ordinary products of integral divisors.

For quotients of elliptic powers, this question is controlled by the valuation pattern of the descended coefficient forms. The main local result of this section says that finite-etale graph slopes force all divided powers of the prescribed graph divisor lattice into ordinary products, in every degree. The proof has two steps: the graph equations produce a symmetric matrix-of-ideals lattice, and a midpoint inequality makes that lattice rank-one generated. Rank-one coefficient divisors have square zero, so their divided powers expand without factorials. The six-axis application supplies the primitive minimal class needed below.

3.1 A local graph theorem

Work over a complete unramified discrete valuation ring O , with uniformizer p . Fix an elliptic curve E , an isogeny $q : E^n \rightarrow Y$ to a principally polarized abelian variety, and extend the p -adic coefficient lattices to O . A symmetric matrix $A \in \text{Sym}_n(O)$ denotes the standard coefficient divisor D_A on E^n , generated by coordinate and Poincare divisors. Define the prescribed graph-divisor lattice and its ordinary product image by

$$\begin{aligned} N_{\text{gr}}(q, O) &= \{D \in \text{NS}(Y) \otimes O : q^*D = D_A \text{ for some } A \in \text{Sym}_n(O)\}, \\ P_{\text{gr}}^k &= \text{im}(\text{Sym}^k N_{\text{gr}}(q, O) \longrightarrow H^{2k}(Y, \mathbf{Z}_p) \otimes O). \end{aligned} \quad (3.1)$$

Because q^* is injective on torsion-free cohomology, we identify $N_{\text{gr}}(q, O)$ with the symmetric matrices whose coefficient divisors descend through q . This is the prescribed lattice even when E has complex multiplication; no extra CM divisor is included.

Suppose the pullback coefficient polarization on the elliptic-power source is

$$G = PB, \quad P = \bigoplus_a p^a I_{M_a}, \quad B = \bigoplus_a B_a, \quad (3.2)$$

where each B_a is unimodular. A *marked finite-etale graph presentation* consists of a symplectic ruling of the primary polarization kernel in which $\ker q$ is the graph of an endomorphism T , with the following properties: T preserves the depth summands M_a ,

$$T^t B^{-1} = B^{-1} T \quad \text{or, equivalently,} \quad B T^t = T B, \quad (3.3)$$

and the algebra generated by T modulo p^a is finite etale on every positive-depth summand. Thus adjointness is taken for the dual coefficient form B^{-1} . No etaleness is required on M_0 .

After a finite unramified extension splitting these etale algebras, the positive-depth summands decompose orthogonally into blocks

$$L_i = L_{a_i, \lambda_i}, \quad T_i = t_i I + p^{a_i} S_i, \quad (3.4)$$

with S_i integral. Indeed, completeness lifts the primitive idempotents of each split finite-etale quotient to idempotents in the algebra generated by T . They are polynomials in T , so (3.3) makes them self-adjoint for B^{-1} . Their images are therefore mutually orthogonal for B^{-1} , and hence for B ; on each image, T is scalar modulo the corresponding power p^{a_i} . We keep the unit part as one block. Write $\delta_{ij} = v_p(t_j - t_i)$, with $v_p(0) = \infty$, for two positive-depth blocks.

Lemma 3.1 (The graph coefficient lattice). *In the split basis, the coefficient lattice of descended graph divisors is the symmetric matrix-of-ideals lattice whose diagonal block i has depth p^{a_i} and whose (i, j) -block has depth*

$$e_{ij} = \max\{a_i, a_j, a_i + a_j - \delta_{ij}\}. \quad (3.5)$$

For the unit block and a positive-depth block j , one has $e_{0j} = a_j$. In particular,

$$e_{ij} \geq \max(a_i, a_j) \geq \left\lceil \frac{a_i + a_j}{2} \right\rceil. \quad (3.6)$$

Proof. Choose the symplectic basis determined by the marked ruling and the depth decomposition. Adjoining the graph generators to the source homology lattice gives the quotient lattice, and in these coordinates the change-of-lattice matrix is

$$C = \begin{pmatrix} P^{-1} & P^{-1}T \\ 0 & I \end{pmatrix}.$$

A coefficient alternating form descends precisely when its matrix transported by C is integral. For a symmetric source coefficient A , direct multiplication gives

$$C \begin{pmatrix} 0 & A \\ -A & 0 \end{pmatrix} C^t = \begin{pmatrix} P^{-1}(AT^t - TA)P^{-1} & P^{-1}A \\ -AP^{-1} & 0 \end{pmatrix}. \quad (3.7)$$

Thus descent is equivalent to the integrality of the three displayed nonzero blocks. Here $A = A^t$ is a symmetric coefficient bilinear form in this normalization. On two positive-depth blocks $L_i \times L_j$, the two off-diagonal blocks first force A_{ij} to have depth $\max(a_i, a_j)$. Substituting $T_i = t_i I + p^{a_i} S_i$ into the upper-left block leaves the additional condition

$$p^{-a_i - a_j} (t_j - t_i) A_{ij} \in M_{r_i \times r_j}(O);$$

the terms containing S_i, S_j are already integral. Intersecting the two conditions gives (3.5). Changing a scalar lift cannot change its maximum. Pairing the unsplit unit block with L_j gives $e_{0j} = a_j$ directly from (3.7), and (3.6) follows. $\square$

The inequality (3.6) is the valuation shadow of the tropical positive semidefinite cone studied by Yu [21]. We need the following signed integral lift of that inequality; in particular, no division by two is permitted.

Lemma 3.2 (Rank-one generation over a DVR). *Let L be a symmetric matrix-of-ideals lattice over O , with diagonal depths p^{a_i} and cross depths $p^{e_{ij}}$. Then L is generated by its rank-one forms if and only if*

$$2e_{ij} \geq a_i + a_j \quad (i \neq j). \quad (3.8)$$

Proof. If $cvv^t \in L$, the two diagonal conditions imply

$$v_p(cv_i v_j) \geq \left\lceil \frac{a_i + a_j}{2} \right\rceil.$$

Hence rank-one forms cannot generate a shallower cross ideal.

Conversely, fix a generator $p^e(uv^t + vu^t)$ of a cross block. Choose $x \geq a_i$ with $y = 2e - x \geq a_j$. The integers x, y have the same parity. If $t = \min(x, y)$, write $x = t + 2r$ and $y = t + 2s$. Then

$$p^t(p^r u + p^s v)(p^r u + p^s v)^t - p^x u u^t - p^y v v^t = p^e(uv^t + vu^t). \quad (3.9)$$

All three rank-one terms lie in L . Together with the diagonal rank-one forms, these identities generate L , without division by two. $\square$

Theorem 3.3 (All-degree graph saturation). *For a marked finite-étale graph presentation, every divided power of a class in its prescribed graph divisor lattice is an ordinary integral divisor product. Equivalently, after localization at p , the divided-power envelope of that lattice and its ordinary product image agree in every degree.*

Proof. After the unramified splitting extension, Lemmas 3.1 and 3.2 express every class in the extended graph divisor lattice as a finite sum $D = R_1 + \cdots + R_m$ of rank-one coefficient classes. These are classes in the coefficient extension of the original lattice, not newly descended line bundles. On the elliptic-power source the alternating form of each R_i has rank two, hence $R_i^2 = 0$. Pullback by the isogeny is injective on torsion-free cohomology, so the same identity holds on the quotient. Therefore

$$\frac{D^k}{k!} = \sum_{1 \leq i_1 < \cdots < i_k \leq m} R_{i_1} \cdots R_{i_k}, \quad (3.10)$$

an O' -linear combination of ordinary products over the splitting ring O' . More precisely,

$$P_{\text{gr}}^k \otimes_O O' = \text{im} \left(\text{Sym}^k(N_{\text{gr}} \otimes_O O') \longrightarrow H^{2k} \otimes_O O' \right).$$

Faithful flatness descends the quotient of these finite lattices, and hence descends (3.10) to O , and therefore to $\mathbf{Z}_p$ in the six-axis application. $\square$

The theorem concerns the divisor lattice selected by the marked elliptic-power presentation. If E has no complex multiplication, this is the full Neron–Severi lattice. At a CM fibre extra divisor classes need not satisfy the graph hypotheses. The statement is cohomological: it neither constructs Chow divided powers nor asserts that every integral Hodge class is a divisor product.

For comparison with the classical rational theory, Milne proves that the Lefschetz classes are the rational algebra generated by divisors [16, Theorem 3.2]. Divided powers in Chow and integral Fourier transforms are studied in [17, 3]. The local result above instead keeps the weighted integral graph lattice and the ordinary, rather than divided, product image.

3.2 The six-axis packet

We now apply the graph theorem to the polarization computed in Section 2.

We use the following elementary depth-one lifting observation. Let U be a unimodular O -lattice with symmetric Gram matrix B . Every endomorphism $\bar{T}$ of U/pU that is self-adjoint for $\bar{B}^{-1}$ has a lift $T \in \text{End}_O(U)$ satisfying

$$T^t B^{-1} = B^{-1} T. \quad (3.11)$$

The construction may be performed separately on orthogonal depth summands. Choose any lift T_0 . Its adjointness defect has the form

$$T_0^t B^{-1} - B^{-1} T_0 = pD,$$

where D is skew-symmetric with zero diagonal. Choose an integral matrix C with $C^t - C = -D$, for example by assigning its entries strictly above and below the diagonal. Then $T = T_0 + pBC$ reduces to $\bar{T}$, and

$$T^t B^{-1} - B^{-1} T = p(D + C^t - C) = 0.$$

No division by two is used. Applying the construction blockwise preserves the depth decomposition.

Lemma 3.4 (Local chart for the six-axis quotient). *Let $G = 6I_5 - J_5$. At each prime $p \mid 6$, the coefficient lattice has an orthogonal decomposition*

$$(\mathbf{Z}_p^5, G) \simeq U_0 \perp pU_1,$$

where U_0 has rank one and U_1 rank four, both unimodular. The principal quotient is trivial on U_0 . On U_1/pU_1 , its graph slope has minimal polynomial $t^2 + t + 1$ at $p = 2$; at $p = 3$ it is a scalar, hence a one-point finite-etale block, after choosing a complementary symplectic ruling if the selected half is vertical. In both cases the slope has a depth-preserving integral lift that is self-adjoint for the dual coefficient form, so these data give a marked finite-etale graph presentation.

Proof. Let $e_1, \dots, e_5$ be the standard basis. Since $\langle e_1, e_1 \rangle = 5$ and $\langle e_1, e_j \rangle = -1$, the vectors

$$v_j = e_j + \frac{1}{5}e_1 \quad (2 \leq j \leq 5)$$

span the orthogonal complement of e_1 over $\mathbf{Z}_p$. Its Gram matrix is $\frac{6}{5}(5I_4 - J_4)$, and $\det(5I_4 - J_4) = 5^3$. This gives the asserted unit line and depth-one block at two and three. The unit line contributes nothing to the p -primary discriminant group, so the p -primary isogeny kernel is supported entirely on U_1/pU_1 .

At two, Proposition 2.3 applies to every smooth fibre and identifies its geometric kernel with one of the two exotic graphs. Isotropy says that either slope is self-adjoint for the reduced dual form. It is a root in $\mathbf{F}_4$ of $t^2 + t + 1$, so its generated algebra is finite etale. The preceding lifting argument supplies the required integral self-adjoint lift on U_1 . At three, the same proposition says that every A_5 -stable half is a scalar graph because the coefficient heart is simple with commutant $\mathbf{F}_3$. More precisely, the four halves form $\mathbf{P}^1(\mathbf{F}_3)$. If the selected member is vertical, choose any other member as the complementary vertical ruling; the selected half then becomes the graph of a scalar (including zero). This change is only in the symplectic ruling of the primary kernel. The chosen symplectic basis of its two-dimensional multiplicity space lifts from $\mathbf{F}_3$ to $\mathbf{Z}_3$: equivalently, reduction $\mathrm{Sp}_2(\mathbf{Z}_3) \rightarrow \mathrm{Sp}_2(\mathbf{F}_3)$ is surjective. Thus it lifts to an integral change of ruling preserving the alternating form; the depth-one coefficient block and its scale $P = 3I$ are unchanged. The algebra generated by the resulting scalar is $\mathbf{F}_3$, hence is a one-point finite-etale block, and the same lifting argument applies. Extend each lift by a scalar on U_0 . No global choice of either slope is needed. $\square$

Theorem 3.5 (Divided powers for the six-axis family). *For every smooth member X_b of the nonstandard A_5 -cubic pencil, the prescribed six-axis graph divisor lattice on J_b has all its integral divided powers in the ordinary divisor-product image. In particular,*

$$\frac{\Theta_b^4}{4!} \in P_{J_b}^4. \quad (3.12)$$

Proof. Only the primes two and three occur in the Smith form $(1, 6, 6, 6, 6)$. Lemma 3.4 verifies the hypotheses of Theorem 3.3 at both primes. At every other prime the isogeny identifies the integral source and target lattices; equivalently the marked presentation has only its unit block $P = I$, so Theorem 3.3 applies trivially (or directly by Lemma 3.2). Thus the image of each desired divided power in the finitely generated quotient $H^{2k}(J_b, \mathbf{Z})/P_{J_b}^k$ vanishes after tensoring with $\mathbf{Z}_p$ for every prime p , and hence vanishes integrally. Finally, Lemma 2.2 gives $f_b^* \Theta_b = \lambda_{\mathcal{A}_b}$. The principal polarization is therefore one of the descended graph divisors defining the prescribed lattice, yielding (3.12). Extra Neron–Severi classes at a special fibre can only enlarge the ordinary product image needed for this last assertion. $\square$

Voisin identifies the primitive minimal class as the exact obstruction needed for cubic threefolds.

Corollary 3.6 (Universal CH_0 -triviality). *Every smooth member of the nonstandard A_5 -cubic pencil is universally CH_0 -trivial.*

Proof. By Theorem 3.5, the primitive minimal class of the intermediate Jacobian is algebraic. Voisin's Corollary 4.4 [20] says that, for a smooth complex cubic threefold, this is equivalent to universal CH_0 -triviality. $\square$

The argument is fibrewise. It constructs neither a horizontal minimal cycle over the pencil nor a relative decomposition of the diagonal. Neither is needed for Corollary 3.6.

4 Framed formal monodromy

The obstruction used here belongs to the small even quantum connection itself. We follow it through each blowup in a weak factorization, without passing to equivalence classes of quantum atoms. This avoids a subtlety: a formal bulk gauge is coefficientwise Laurent in the loop coordinate, but need not specialize to a meromorphic gauge at a nearby analytic point.

The proof has three stages. We first define the framed multiplicity ν_6 . We then prove its projective-bundle and blowup formulas, using divisor tags when the center Novikov map is not injective. Finally we show that centers of dimension at most two have $\nu_6 = 0$, so weak factorization makes ν_6 birationally invariant through dimension four.

Let T be a smooth projective variety. Write

$$\Lambda_T = \mathbf{C}[[\mathrm{NE}_{\mathrm{num}}^{\mathbf{Z}}(T)]]$$

for the completed monoid ring of the image of effective homology classes in the numerical curve lattice

$$N_1(T) = \mathrm{im}(H_2(T, \mathbf{Z}) \longrightarrow \mathrm{Hom}(\mathrm{NS}(T)/\mathrm{tors}, \mathbf{Z})).$$

Thus $N_1(T)$ is torsion-free and is separated by integral divisors. Let $\mathcal{K}_T$ be an algebraic closure of its fraction field. The loop coordinate z remains independent: in particular, $\mathcal{K}_T$ contains no chosen root of z . We first isolate the cyclotomic part of formal monodromy without choosing values for the exponentials of arbitrary elements of $\mathcal{K}_T$.

Let $K \supset \mathbf{C}$ be an algebraically closed field of characteristic zero. Choose a complement $K = \mathbf{C} \oplus V$ of $\mathbf{Q}$ -vector spaces, let $K[V]$ be the group algebra of the additive group of V , and let Ω_V be an algebraic closure of its fraction field. Thus $K \subset \Omega_V$. The map

$$\mathrm{Exp}_V(c + v) = e^{2\pi i c}[v]: (K, +) \longrightarrow \Omega_V^\times \quad (4.0)$$

is a group homomorphism. Since V is torsion-free, $K[V]$ is a domain, and its distinct monomials are linearly independent. Consequently

$$\ker(\mathrm{Exp}_V) = \mathbf{Z}, \quad \mathrm{Exp}_V(a) \text{ is a root of unity} \iff a \in \mathbf{Q}. \quad (4.0a)$$

Apply Levelt–Turrittin after a ramification $z = w^e$. In the formal decomposition, write the residue on a regular-singular block as $R = R_s + R_n$, with commuting semisimple and nilpotent parts. In the universal formal solution algebra, define a full turn of w on that block by

$$M_{\mathrm{RS}, V} = \mathrm{Exp}_V(R_s) \exp(2\pi i R_n),$$

where Exp_V is applied to the eigenvalues of R_s . The lift of one turn of the original z -disc acts on the formal symbols by $w \mapsto e^{2\pi i/e} w$, and on the blocks by their descent intertwiners. Its e -th power is $M_{\text{RS},V}$ on every returned regular-singular block. An integral change of residue representative multiplies the formal solution by an integral power of w , so it conjugates this lifted action rather than changing it. Since the coefficient and descent matrices lie in $K \subset \Omega_V$, this defines the framed operator $M_{\text{f},V}$ over Ω_V . Ramified exponential factors are thereby permuted rather than made single-valued by changing the frame.

Choice independence. For every root of unity $\zeta \in \mathbf{C}$, its algebraic multiplicity as an eigenvalue of $M_{\text{f},V}$ is independent of the complement V , the Levelt–Turrittin ramification and decomposition, and the algebraic closure Ω_V . It is also unchanged when the chosen algebraic closure of the original coefficient field is replaced by a common algebraically closed overfield. When $K = \mathbf{C}$, it is the usual multiplicity in the analytic formal monodromy on the original z -disc. We use here the functoriality and uniqueness of the ramified formal decomposition; see Sabbah’s treatment of ramified descent and the functorial Levelt–Turrittin decomposition [19, Lecture 5, Section 5.c, proofs of Theorem 5.8 and Proposition 5.10]. Consider one orbit of Levelt–Turrittin blocks under a turn of the original disc, let d be its length, and write $A_i : E_i \rightarrow E_{i+1}$ for its successive deck/descent maps, with indices modulo d . Put $U = A_{d-1} \cdots A_0$ on E_0 . A cyclic-block determinant gives

$$\chi_{\bigoplus_i E_i}(X) = \det(X^d I - U). \quad (4.0b)$$

Put $r = e/d$. Equivariance of the descent datum gives the intrinsic return relation

$$U^r = M_{\text{RS},V}. \quad (4.0c)$$

Thus, if μ is an eigenvalue of U , then $\mu^r = \text{Exp}_V(\rho)$ for a residue exponent ρ , with algebraic multiplicities read on the corresponding generalized eigenspaces. If a root of unity ζ occurs in (4.0b), then $\mu = \zeta^d$ and $\text{Exp}_V(\rho) = \zeta^e$; hence $\rho \in \mathbf{Q}$ by (4.0a). Conversely, if ρ is rational, then μ^r is torsion, so every corresponding μ , and hence every root of $X^d - \mu$, is torsion. On these rational residue classes Exp_V is the usual complex exponential; the finite descent operator is the same intrinsic operator. The exact roots of unity and their algebraic multiplicities are therefore those of the ordinary complex construction. Nonrational residue classes contribute no root of unity. This criterion does not depend on V and does not require a fractional residue factor to commute with the descent action.

For completeness, compare two Levelt–Turrittin choices after pulling both to a common ramification $z = u^E$. Equivariant uniqueness of the formal decomposition identifies their total horizontal solution spaces and carries the original- z -turn deck/descent operator for one choice to the scalar extension of the other. Thus the product of all orbit polynomials (4.0b), rather than any individually labelled return map, is invariant under refinement and re-decomposition. The same argument after a common algebraically closed coefficient extension preserves the intrinsic multiset of residue classes and finite descent characters. The relation (4.0c) is itself invariant under integral changes of residue representative. Hence the cyclotomic test above and its algebraic multiplicities depend only on the original framed module, not on the coefficient closure or complement. Finally, for $K = \mathbf{C}$ one has $V = 0$ and $\text{Exp}_V(a) = e^{2\pi i a}$, which gives the usual analytic formal monodromy.

Definition 4.1. Apply the preceding construction to the small even quantum connection over $\mathcal{K}_T((z))$, and put

$$\nu_6(T) = \text{mult}_{e^{\pi i/3}} M_{\text{f},V}(T) + \text{mult}_{e^{-\pi i/3}} M_{\text{f},V}(T).$$

The multiplicities are algebraic multiplicities in the characteristic polynomial. The choice-independence argument above makes their sum independent of all the auxiliary choices.

The frame matters. Adjoining roots of Novikov monomials is an algebraic coefficient extension and is harmless by choice independence. Adjoining a root of z , by contrast, would replace the original loop and could erase the deck part of the operator.

Definition 4.2 (Pro-Laurent gauge group). Let B be a complete separated multiplicatively filtered k -algebra, and let V be a finite-dimensional k -vector space. The pro-Laurent gauge group is

$$\mathcal{G}_{B,F}^{\text{pro}}(V) = \varprojlim_N \text{GL}(V \otimes (B/F^N B)((z))).$$

It is generally larger than $\text{GL}(V \otimes B((z)))$, because the lower Laurent bound may decrease with N . This definition packages only compatible invertible matrices; it does not posit Levelt–Turrittin theory or an independently defined monodromy operator over the nonfield quotients $B/F^N B$.

We first record two formal facts used in the comparison theorems. They also fix the completion convention in the proof.

Lemma 4.3 (Coefficientwise base change). *Let A be a $\mathbf{C}$ -domain, let K be an algebraically closed overfield containing $\text{Frac}(A)$, and let Ω_V be the universal exponential field used above after a formal differential module D has been scalar-extended to $K((z))$. Let B be a complete separated multiplicatively filtered A -algebra and put*

$$\mathcal{L}_{B,F} = \varprojlim_N (B/F^N B)((z)).$$

Assume that the canonical map $A \rightarrow \mathcal{L}_{B,F}$ is injective. Then

$$\widetilde{\mathcal{L}}_{B,F} = \Omega_V \otimes_{\text{Frac}(A)} (\text{Frac}(A) \otimes_A \mathcal{L}_{B,F}) \quad (4.0d)$$

is a nonzero coefficient algebra containing the framed operator $M_{f,V}(D)$ and receiving the image of every pro-Laurent gauge over B . Conjugation by the image of such a gauge preserves its characteristic polynomial. If the gauge uses only integral powers of z at every finite level, this conjugacy preserves the original- z -disc frame.

Proof. Injectivity makes $\text{Frac}(A) \otimes_A \mathcal{L}_{B,F}$ a nonzero $\text{Frac}(A)$ -algebra; scalar extension by the field Ω_V is faithful. The canonical ring map sends a compatible gauge family and its inverse to an invertible pair over $\widetilde{\mathcal{L}}_{B,F}$. Extend $M_{f,V}(D)$ to the same matrix algebra and conjugate there. The determinant identity $\det(X - GMG^{-1}) = \det(X - M)$ holds over every commutative coefficient ring. Finally, the deck action fixes every integral power of z at each finite level, so the compatible gauge is single-valued for the framed turn. $\square$

Remark 4.4 (What the inverse limit requires). Concretely, an element of $\mathcal{L}_{B,F}$ is a series $\sum_{k \in \mathbf{Z}} c_k z^k$ with $c_k \in B$ such that, for every N , all but finitely many $k < 0$ have $c_k \in F^N B$; products converge coefficientwise because the filtration is multiplicative. Since V is finite-dimensional, the inverse limit commutes with the matrix construction, so

$$\mathcal{G}_{B,F}^{\text{pro}}(V) = \text{GL}(V \otimes \mathcal{L}_{B,F}),$$

and a compatible family has its inverse in the same group. No uniform Laurent bound is needed at any point: the lower bound at level N may decrease without limit in N while the family itself is one invertible matrix over one ring. This is the only role the inverse limit plays below.

Lemma 4.5 (Formal base shift). *Let A be a $\mathbf{C}$ -domain and let B be a complete separated multiplicatively filtered A -algebra whose pro-Laurent ring $\mathcal{L}_{B,F}$ contains A injectively. Suppose a pullback of the even quantum connection of T is defined over B , its small connection is defined over $\text{Frac}(A)$, and its bulk coordinate has the form*

$$a_0 \mathbf{1} + a_2^\circ + \eta, \quad a_2^\circ \in H^2(T) \otimes F^0 B, \quad \eta \in F^1 B \hat{\otimes} H^{\text{ev}}(T).$$

Assume that a_0 and a_2° are constant in the positive filtration. Assume also that the units $e^{\langle a_2^\circ, d \rangle} \in A^\times$ are defined for numerical curve classes d , form a multiplicative character, and hence define the displayed coefficient automorphism of A , extending continuously to B . Then the positive-filtration bulk equations have unique normalized gauges G_N over every $B/F^N B$, compatible under reduction. Their inverse limit G belongs to $\mathcal{G}_{B,F}^{\text{pro}}(H^{\text{ev}}(T))$. After extending coefficients as in (4.0d), transporting the small framed operator through G , after the divisor substitution

$$Q^d \mapsto e^{\langle a_2^\circ, d \rangle} Q^d \tag{4.1}$$

defines a comparison-transported framed operator. Its characteristic polynomial is equal to the divisor-substituted small framed characteristic polynomial in $\mathcal{L}_{B,F}[X]$.

Proof. The string equation separates the rank-one exponential twist $\exp(-a_0/z)$. When a_0 is not filtration-nilpotent this twist need not belong to the pro-Laurent gauge group, but a full turn on the original z -disc fixes it, so its framed monodromy is trivial. The pro-Laurent gauge below is used only for the positive-filtration term η . The divisor equation gives (4.1). It remains to remove η , including its component in $H^2(T)$.

Modulo $F^N B$, the flat bulk equations

$$\partial_{\eta_i} G = -z^{-1}(\phi_i \star_\eta) G, \quad G|_{\eta=0} = 1,$$

are compatible because the even quantum connection is flat: the mixed derivatives of the quantum-product matrices agree, and those matrices commute by commutativity and associativity. To derive the Laurent bound, write $G = \sum_\alpha G_\alpha \eta^\alpha$. The coefficient of η^α in the i -th equation is

$$(\alpha_i + 1)G_{\alpha+e_i} = -z^{-1}[(\phi_i \star_\eta)G]_\alpha. \tag{4.1a}$$

The positive integer $\alpha_i + 1$ is invertible over $\mathbf{C}$. Strong induction on $|\alpha|$, with flatness making the choices of i consistent, gives the unique normalized solution; G_α contains no power of z below $-|\alpha|$. Full flatness also makes the G -gauged z -connection independent of η ; at $\eta = 0$ it is the small connection, so G is the required connection gauge. After substituting η , a term of total bulk degree m lies in $F^m B$; modulo $F^N B$, only the finitely many degrees $m < N$ survive. The resulting finite sum is congruent to the identity modulo $F^1 B$, hence is invertible, and has a Laurent lower bound at least $-(N-1)$. That bound decreases with N , which costs nothing by Remark 4.4: the transported operator is formed over $\mathcal{L}_{B,F}$ and never over $B((z))$. Uniqueness makes these solutions compatible in N . Their inverse limit belongs to

$$\mathcal{G}_{B,F}^{\text{pro}}(H^{\text{ev}}(T)). \tag{4.1b}$$

Write G_N for the resulting normalized gauge over $B/F^N B$. Uniqueness gives $G_{N+1} \bmod F^N = G_N$, so the family defines $G \in \mathcal{G}_{B,F}^{\text{pro}}$. No framed operator is base-changed to an individual nonfield quotient. Instead pass first to the inverse-limit ring $\mathcal{L}_{B,F}$, then localize its A -coefficients and extend them to Ω_V as in (4.0d). In that single coefficient algebra define

$$M^{\text{bulk}} = G M_{f,V}^{\text{small,shifted}} G^{-1}. \tag{4.1c}$$

Every G_N uses only integral powers of z , so the compatible gauge is single-valued on the original disc. Lemma 4.3 gives the asserted characteristic-polynomial equality. No Levelt–Turrittin decomposition or framed operator over a nonfield quotient, and no evaluation at a nonnilpotent analytic bulk point, is used. $\square$

We now apply Iritani’s quantum blowup theorem and the Iritani–Koto projective-bundle theorem. Both are stated over completed homological Novikov rings. Iritani’s subsequent notes record explicitly that the blowup change of variables and decomposition preserve cohomological parity [12, Section 2].

Lemma 4.6 (Numerical Novikov base change). *The quotient from the effective homology monoid to its image in $N_1(T)$ extends continuously to completed monoid rings, and the quantum connection and the two comparison theorems base-change along it.*

Proof. Filter both monoids by intersection with an ample divisor. Effective homology classes below a fixed cutoff form a finite set, as follows from the finite-type Chow varieties of cycles of bounded degree. The quotient is therefore defined on every truncated monoid ring and passes to the inverse limit. Explicitly, the coefficient of $Q^{\vec{d}}$ in the image of $\sum_d a_d Q^d$ is

$$\sum_{d \rightarrow \vec{d}} a_d.$$

The sum is finite at every cutoff, and additivity of the quotient makes this coefficient-wise pushforward a unital homomorphism for completed convolution. Thus the quantum products descend by grouping Gromov–Witten coefficients with the same numerical class. Pushforward of curves, pairing with $c_1(N_{C/T})$, and the derivations $Q\partial_Q$ are constant on these fibres. At each cutoff, apply the resulting ring homomorphism to the comparison map and to its stated inverse. Scalar extension preserves their two inverse identities. Grouping coefficients identifies the scalar-extended quantum module with the numerical quantum module, and the preceding fiberwise constancy identifies its connection operators and divisor substitutions. Passing through the compatible cutoffs proves the asserted completed base change. $\square$

Proposition 4.7 (Framed operation formulas). *Let $V \rightarrow T$ be a vector bundle of rank $r \geq 2$. Then*

$$\nu_6(\mathbf{P}_T(V)) = r\nu_6(T). \quad (4.2)$$

Let $\tilde{T} = \mathrm{Bl}_C T$, where $C \subset T$ is smooth of codimension $c \geq 2$. In Iritani’s comparison field one has

$$\nu_6(\tilde{T}) = \nu_6(T) + \sum_{j=0}^{c-2} \nu_6(C; \chi_j), \quad (4.3)$$

where χ_j is the numerical Novikov specialization of the j -th center summand. The endpoint terms in (4.3) are the intrinsic small invariants; the center terms are computed after the displayed specializations.

Proof. For each of the finitely many endpoint and summand comparisons, use the faithful graded Laurent receiver constructed below. Their coefficient exponent lattices embed in one ordered direct-sum lattice. Take an algebraic closure of that coefficient Hahn field, choose one complement to $\mathbf{C}$, and form its universal exponential field. Only afterward adjoin the distinguished integral exponent of the independent loop coordinate z . Choice independence shows that this common choice does not alter any cyclotomic multiplicity.

Iritani's Theorem 5.18 gives a formal coordinate isomorphism and a quantum- D -module isomorphism

$$\mathrm{QDM}(\tilde{T}) \cong \mathrm{QDM}(T) \oplus \mathrm{QDM}(C)^{\oplus(c-1)}$$

over a Laurent extension in the exceptional Novikov variable [11, Theorem 5.18]. The isomorphism and its inverse use only integral powers of z , and the mirror coordinates are independent of z . The comparison field adjoins q^{-1/s_c} , where $s_c = c - 1$ for even c and $s_c = 2(c - 1)$ for odd c ; this is only a coefficient extension. The reconstruction coordinate below is $u = q^{-1/(c-1)}$. The center base shift is controlled before passage to the final Laurent chart. Put $u = q^{-1/(c-1)}$ and use the reconstruction coordinates of [11, Section 5.8.2]:

$$\varsigma_j = \varsigma_j^\circ + s_j, \quad \varsigma_j^\circ = -(c-1)\lambda_j + h_{C,j} + O(u). \quad (4.4)$$

For the j -th target summand, let R_j be the image of $Q_C^d \mapsto Q^{i_*d} u^{\rho_C \cdot d}$, together with u and s_j , and set

$$\begin{aligned} J_j &= (u, \{Q^{i_*d} u^{\rho_C \cdot d} : d \neq 0\}, \text{components of } s_j) \subset R_j, \\ B_j &= \varprojlim_N R_j / J_j^N, & F^N B_j &= \ker(B_j \rightarrow R_j / J_j^N). \end{aligned}$$

Here R_j is a subring of the Laurent monoid domain occurring in the comparison chart, and B_j is, by construction, complete and separated for the multiplicative inverse-limit filtration $F^\bullet B_j$. Reduction $B_j / F^{N+1} B_j \rightarrow B_j / F^N B_j$ preserves the connection formulas. To make separation explicit, fix an ample divisor H on T . On the effective cone of C , the linear functional $d \mapsto \rho_C \cdot d$ is bounded in absolute value by a constant times $H \cdot i_* d$. Choose an integer L so large that

$$w(Q^{i_*d} u^{\rho_C \cdot d}) = L(H \cdot i_* d) + \rho_C \cdot d \geq 1 \quad (d \neq 0),$$

and set $w(u) = 1$ and $w(s_{j,\ell}) = 1$ for every component of s_j . This additive integer weight is defined on the ambient Laurent monomial lattice, so it respects every relation in the image monoid defining R_j . Every generator of J_j has weight at least one, hence every monomial in J_j^N has weight at least N . The Novikov finite-below support condition gives every nonzero element of R_j a finite lowest weight. Consequently $\bigcap_N J_j^N = 0$, so the canonical maps from R_j into B_j and then into $\mathcal{L}_{B_j, F}$ are injective. The center connection is defined over this image ring by [11, Remark 5.6]. Formula (4.4), together with (5.45), (5.47), and the initial asymptotics (5.27)–(5.30), shows that after the unit term $-(c-1)\lambda_j$ and fixed divisor $h_{C,j}$ are removed, its target bulk coordinate lies in $J_j H^*(C)$. The noninjective center monomial map merely replaces the source ring by its image.

First separate the single-valued rank-one exponential twist associated with $-(c-1)\lambda_j$, and apply the fixed divisor shift. The normalized connection descends to B_j . Apply Lemma 4.5 with $A = R_j$. It constructs the compatible gauges first and compares the framed operators only after forming $\mathcal{L}_{B_j, F}$, localizing its R_j -coefficients, and extending to a universal exponential field. No framed operator is placed over an individual quotient.

Here is the required continuity check without extending J_j to a ring in which u is a unit. Put $v = q^{-1/s_c}$, so $v^e = u$ for $e = s_c/(c-1) \in \{1, 2\}$. Iritani's nonlocalized homogeneous variables include

$$v, \quad Q^\delta, \quad q^{k/(c-1)} \theta_{Z,k} = u^{-k} \theta_{Z,k}, \quad (4.4a)$$

with the analogous scaled components on the ambient summand. The filtration F_N in the proof of [11, Proposition 5.4, especially (5.13)–(5.14)] orders these monomials first by ample Novikov degree and then by bulk weight; homogeneity controls the possible negative

v -power. It has finite-below support and is multiplicative. This is precisely the graded meaning of $\mathbf{C}[z]((v))[[Q, \theta]]$ stipulated in [11, Remarks 1.3–1.5], rather than an ordinary J_j -adic completion after v has been inverted.

The coefficient identification is the full nonlinear coordinate map, not merely the monomial substitution above. Namely,

$$\tilde{\tau} \longmapsto (\tau(\tilde{\tau}), \varsigma_0(\tilde{\tau}), \dots, \varsigma_{c-2}(\tilde{\tau})) \quad (4.4b)$$

is the graded formal coordinate isomorphism of [11, Corollary 1.2 and Theorem 5.18]. Pull each target completion B_j , its connection, and its normalized gauge through (4.4b) before embedding it. This pullback and the inverse coordinate change preserve finite-below support: substitution of positive-order homogeneous series has only finitely many contributions in each F_N -degree, and the same induction on degree constructs the inverse because the displayed coordinate map has invertible linear term.

Let Γ_j^{coeff} be the coefficient exponent lattice of these source homogeneous variables, including v , the source Novikov monomials, and the scaled source bulk variables. The pulled-back monomials of R_j are elements of its graded completion. Order this lattice by the preceding positive multidegree, then by any lexicographic refinement. The finite-below property makes every permitted support well ordered. The coefficient of a fixed monomial in a product is a finite sum, and a nonzero series retains its least supported monomial; Laurent localization only translates supports. Coefficientwise inclusion therefore embeds faithfully both the pulled-back J_j -complete gauge ring and Iritani's graded Laurent comparison ring into $H_{0,j} = \mathbf{C}((\Gamma_j^{\text{coeff}}))$. Take an algebraic closure of $H_{0,j}$ and its universal exponential field $\Omega_{V,j}$ as in Definition 4.1, using a new independent loop coordinate z . The numerical small module is first scalar-extended along its coefficient embedding into $H_{0,j}$; choice independence preserves its cyclotomic multiplicities. Now put

$$H_{\text{tot},j} = \mathbf{C}((\Gamma_j^{\text{coeff}} \oplus \mathbf{Z}e_z)), \quad \mathcal{R}_j = \Omega_{V,j} \otimes_{H_{0,j}} H_{\text{tot},j}, \quad (4.4c)$$

where e_z is ordered after the positive coefficient degree and maps to that same distinguished loop coordinate. Both factors are field extensions of $H_{0,j}$, so $\mathcal{R}_j \neq 0$ and each factor injects. The derivation and framed turn are the ones fixed on this independent z before tensoring; they fix $\Omega_{V,j}$ and the coefficient Hahn field, and no root of z is adjoined. The recurrence bound (4.1a) puts the compatible gauge and its inverse in $\mathcal{R}_j$. Theorem 5.18 places Ψ , Ψ^{-1} , and their two inverse identities in the same graded Laurent ring. Hence all comparison maps, gauges, and framed operators coexist in one nonzero receiver $\mathcal{R}_j$, with the source and target monomials identified by the displayed coordinate map. No uniform u -denominator is asserted, and no adic ideal is extended across Laurent inversion. Theorem 5.18 supplies both maps with only integral powers of z , so they preserve the framed characteristic polynomial on the original disc. The fixed-divisor substitution is a z -constant coefficient automorphism, so it does not change ν_6 ; it is well defined on the image because $h_{C,j}$ is a scalar multiple of ρ_C . Thus each center target summand has the framed monodromy of its specialized small connection. The ambient target uses $\tau = \tau^\circ + t$, with $\tau^\circ = O(q^{-1})$, and the same finite-quotient argument. Direct-sum additivity gives (4.3).

The endpoint map is faithful: on numerical curve lattices

$$\bar{\beta} \longmapsto (\phi_* \bar{\beta}, -E \cdot \bar{\beta})$$

is injective because $N^1(\tilde{T}) = \phi^* N^1(T) \oplus \mathbf{Z}[E]$. Hence the Laurent extension preserves the intrinsic endpoint characteristic polynomials.

The projective-bundle proof is parallel, again in reconstruction coordinates. Its comparison field adjoins $q^{-1/r'}$, with $r' = r$ when $r - 1$ is even and $r' = 2r$ when $r - 1$ is odd; the

reconstruction coordinate remains $u = q^{-1/r}$. The extra root is again a coefficient extension. Put $u = q^{-1/r}$. For its j -th target summand let R_j be the image of $Q_T^d \mapsto Q^d u^{c_1(V) \cdot d}$, together with u and s_j , and set

$$J_j = (u, \{Q^d u^{c_1(V) \cdot d} : d \neq 0\}, \text{components of } s_j) \subset R_j,$$

$$B_j = \varprojlim_N R_j / J_j^N, \quad F^N B_j = \ker(B_j \rightarrow R_j / J_j^N).$$

Again R_j is a subring of the relevant Laurent monoid domain and B_j is complete and separated for its multiplicative inverse-limit filtration $F^\bullet B_j$. Fix an ample divisor H on T . The linear functional $d \mapsto c_1(V) \cdot d$ is bounded in absolute value by a constant times $H \cdot d$ on the effective cone. Choose an integer L so large that

$$w(Q^d u^{c_1(V) \cdot d}) = L(H \cdot d) + c_1(V) \cdot d \geq 1 \quad (d \neq 0),$$

and put $w(u) = w(s_{j,\ell}) = 1$ for every component of s_j . This is an additive integer weight on the ambient Laurent monomial lattice and therefore respects all relations in the image monoid. Thus J_j^N has weight at least N , while every nonzero Novikov series in R_j has a finite lowest weight. It follows that $\bigcap_N J_j^N = 0$, hence there are injective maps $R_j \rightarrow B_j \rightarrow \mathcal{L}_{B_j, F}$. Proposition 5.6 of [13] puts the normalized target coordinate in the maximal ideal, hence in $J_j H^{\text{ev}}(T)$, and the displayed exponent shows that the fixed-divisor substitution factors through the image monoid. Theorem 5.1(4) and Proposition 5.6 of [13] give

$$\varsigma_j = \varsigma_j^\circ + s_j, \quad \varsigma_j^\circ = r\lambda_j - \frac{2\pi i j}{r} c_1(V) + O(u).$$

Here s_j collects the bulk directions and ς_j° is the value at the origin. Separate the Laurent unit as a single-valued rank-one exponential twist and apply the fixed divisor shift first. The normalized connection descends to B_j ; apply Lemma 4.5 with $A = R_j$. It produces the compatible pro-Laurent gauge and compares framed operators only after the inverse limit, coefficient localization, and universal-exponential extension. For completeness, put $v = q^{-1/r'}$, so $v^e = u$ for $e = r'/r \in \{1, 2\}$. The nonlocalized coefficient ring of [13, (5.3)] is a power series ring in the homogeneous generators

$$v, \quad q^{-c_1(V) \cdot d/r} Q^d, \quad q^{k/r} \widehat{\tau}_{i,k} = u^{-k} \widehat{\tau}_{i,k}. \quad (4.4d)$$

Its exponent monoid is generated by these variables, finitely many because $0 \leq k \leq r-1$, and the order that ranks ample Novikov degree first and bulk weight second is positive on each of them. Every element therefore has well-ordered support: this is the finite-below property used below, and it is the graded-completion convention of [13, Section 5.1, especially (5.1)–(5.3)]. The grading controls the negative v -power. The map

$$\widehat{\tau} \mapsto (\varsigma_0(\widehat{\tau}), \dots, \varsigma_{r-1}(\widehat{\tau})) \quad (4.4e)$$

has invertible Jacobian by Theorem 5.1(5) of [13], an item corrected in the fourth version of that preprint, dated 31 January 2026, and hence is a graded formal coordinate isomorphism; its derivative gives the ring isomorphism in [13, Corollary 1.8]. Pull the target completions, connections, and gauges through this full map. The map and its inverse are continuous for the homogeneous finite-below topology: at each degree substitution is a finite sum, and the inverse is constructed recursively from the invertible linear term.

Form the coefficient exponent lattice Γ_j^{coeff} from the source homogeneous generators in (4.4d), and refine this multidegree to a total order. The same finite-below support argument embeds the gauge and comparison coefficient rings in its Hahn field $H_{0,j} = \mathbf{C}((\Gamma_j^{\text{coeff}}))$. Take

its algebraic closure and universal exponential field $\Omega_{V,j}$ before adjoining e_z , and use the nonzero receiver

$$\Omega_{V,j} \otimes_{H_{0,j}} \mathbf{C}((\Gamma_j^{\text{coeff}} \oplus \mathbf{Z}e_z)).$$

Theorem 5.1 and Proposition 5.6 put the comparison maps, their inverses, and the connection identities in that same receiver. Thus growing negative u -order coupled to positive bulk degree is retained rather than replaced by a fictitious uniform denominator. Since the comparison maps use integral powers of z , the framed characteristic polynomial is preserved. Again the divisor substitution is a z -constant coefficient automorphism; it factors through the displayed u -exponent. This gives r copies of the intrinsic base module and proves (4.2). The base Novikov map is the embedding in [13, (1.1) and Remark 5.2]. The theorem assumes $V^\vee$ globally generated. Tensoring V by a sufficiently negative line bundle achieves this without changing $\mathbf{P}_T(V)$ [13, Remark 1.2]; the resulting divisor shift preserves framed monodromy by Lemma 4.5. All other Novikov extensions are in the coefficient direction. After the odd bulk variables vanish, pullback, pushforward, Fourier transformation, and multiplication by the characteristic-class factors all preserve cohomological parity. Hence both decompositions restrict to the even quantum connections used here. Formula (4.2) is tautological when $r = 1$. $\square$

The center map in (4.3) is not necessarily injective [11, (5.15)]. Thus generic vanishing for C does not by itself imply that its specialized term vanishes. The next lemma supplies the missing implication.

Definition 4.8. A monomial map $\chi : \Lambda_T \rightarrow A$ is *strictly Novikov-admissible* if A is a complete separated valued domain with domain associated graded, every monomial has nonzero image, and

$$v(\chi(Q^d)) = \ell(d), \quad (4.5)$$

where ℓ is positive and proper on the effective numerical monoid.

We write $\nu_6(T; \chi)$ for the framed small-even invariant after scalar extension along such a map.

The center maps in (4.3) are strictly admissible. Give the exceptional Laurent variable valuation zero and use an ample divisor H on T . Then

$$v_H(Q^{i_*d} q^{-\rho_C \cdot d / (c-1)}) = (H|_C) \cdot d.$$

The restriction $H|_C$ is ample, which gives positivity and properness; the associated graded Laurent monoid ring is a domain.

Lemma 4.9 (Divisor tagging). *Let $\chi : \Lambda_T \rightarrow A$ be strictly Novikov-admissible. If the intrinsic small even quantum connection of T has no primitive-sixth framed monodromy, neither does its specialization along χ .*

Proof. If $N_1(T) = 0$, then the numerical effective monoid is $\{0\}$, so the coefficient map is already injective and there is nothing to tag. Assume otherwise. Choose integral divisors $D_1, \dots, D_\rho$ whose pairings separate $N_1(T)$, and introduce formal variables $\mathbf{t} = (t_1, \dots, t_\rho)$. Define

$$\chi_{\mathbf{t}}(Q^d) = \chi(Q^d) \exp\left(\sum_i t_i (D_i \cdot d)\right). \quad (4.6)$$

This map from Λ_T to $A[[\mathbf{t}]]$ is injective. To prove this, take $f = \sum_d c_d Q^d \neq 0$ and put

$$\mu = \min\{v(\chi(Q^d)) : c_d \neq 0\} = \min\{\ell(d) : c_d \neq 0\}, \quad S_\mu = \{d : c_d \neq 0, \ell(d) = \mu\}.$$

Properness in (4.5) makes S_μ finite and nonempty. Write in_v for the initial class in the associated graded. The degree- μ initial form is

$$\text{in}_\mu(\chi_t(f)) = \sum_{d \in S_\mu} c_d \text{in}_v(\chi(Q^d)) \exp\left(\sum_i t_i(D_i \cdot d)\right) \in \text{gr}_v(A)[[t]],$$

and every displayed summand coefficient is nonzero. Every term outside S_μ has valuation strictly greater than μ , so a nonzero initial form proves $\chi_t(f) \neq 0$. This is a finite combination of distinct exponential characters. The finitely many exponent vectors are distinct integral vectors. Choose $a = (a_i) \in \mathbf{Z}^\rho$ outside the finitely many integral hyperplanes on which two dot products agree, and substitute $t_i = a_i s$. The resulting exponents are distinct integers. Their derivatives at $s = 0$ form a Vandermonde matrix with nonzero integer determinant, which remains invertible in the characteristic-zero fraction field. The characters are therefore linearly independent.

The domain hypothesis on $\text{gr}_v(A)$ enters exactly here. It makes v multiplicative on A , so each monomial image has valuation exactly $\ell(d)$, the exponential factor is a unit of valuation zero, and no product of monomial images jumps in valuation. Terms with $\ell(d) > \mu$ therefore contribute nothing in degree μ , and each $d \in S_\mu$ contributes its initial form with c_d intact. Were the associated graded to have zero divisors, v would be a filtration rather than a valuation, and the displayed degree- μ identity could fail.

Let $k_\chi = \overline{\text{Frac } A}$. After embedding the two fraction fields in a common algebraic closure, injectivity makes the χ_t -module a scalar extension of the intrinsic module. On the other hand, the divisor equation identifies (4.6) with the degree-two formal bulk pullback of the χ -module over $B = k_\chi[[t]]$. Apply the finite-quotient argument in Lemma 4.5 with $\eta = \sum_i t_i D_i$: its integral- z bulk gauge removes the positive divisor component. Hence vanishing over the intrinsic field implies vanishing after χ . $\square$

Proposition 4.10 (Low-dimensional vanishing). *Let T be a point, a smooth projective curve, or a smooth projective surface. For every strictly Novikov-admissible specialization χ ,*

$$\nu_6(T; \chi) = 0. \quad (4.7)$$

Proof. We first prove the intrinsic generic statement. If K_T is nef, the proof of [14, Claim 6.15] applies coefficient by coefficient over the small numerical Novikov ring. Set

$$\text{Gr} = \frac{\text{Deg} - \dim T}{2}, \quad g = \text{Gr} + \frac{1}{2}E,$$

where E is the identity in cohomological degrees of parity opposite to $\dim T$ and is zero in the other degrees. The operator g has integral eigenvalues. Coefficientwise in Q^β , multiplication by $c_1(T)$ sends

$$H^i(T) \longrightarrow H^{i+2-2c_1(T) \cdot \beta}(T).$$

If K_T is nef, then $c_1(T) \cdot \beta \leq 0$, so the Gr-weight rises by $1 - c_1(T) \cdot \beta \geq 1$. Conjugation by z^g therefore turns $\nabla + \frac{1}{2}E dz/z$ into a regular-singular connection with nilpotent residue. Undoing the half-parity correction permits only the monodromy eigenvalues 1 and -1 : quantum multiplication preserves cohomological parity, so the parity operator commutes with the transformed nilpotent residue. A primitive sixth root cannot occur.

This treats curves of positive genus and minimal surfaces with nef canonical class. A point is immediate. The remaining minimal varieties are $\mathbf{P}^1$, $\mathbf{P}^2$, and ruled surfaces; (4.2) reduces them to a point or a curve. Every nonminimal surface is obtained from a minimal surface by point blowups, and (4.3) adds only point summands. Hence the intrinsic generic invariant vanishes in every case. Lemma 4.9 gives (4.7). $\square$

Theorem 4.11 (Birational invariance through dimension four). *If Y and Y' are birational smooth projective complex varieties of the same dimension at most four, then*

$$\nu_6(Y) = \nu_6(Y'). \quad (4.8)$$

Consequently, for smooth projective threefolds,

$$Y \times \mathbf{P}^1 \dashrightarrow Y' \times \mathbf{P}^1 \implies \nu_6(Y) = \nu_6(Y').$$

Proof. Weak factorization decomposes a birational map between smooth varieties into blowups and blowdowns with smooth centers of codimension at least two [1]. In dimension at most four every nontrivial center has dimension at most two. The calculation following Definition 4.8 shows that every center specialization in (4.3) is strictly admissible. Proposition 4.10 therefore makes every center term zero, so the endpoint values agree at each step.

If Y, Y' are smooth threefolds and $Y \times \mathbf{P}^1 \dashrightarrow Y' \times \mathbf{P}^1$, apply the first assertion in dimension four and (4.2):

$$2\nu_6(Y) = \nu_6(Y \times \mathbf{P}^1) = \nu_6(Y' \times \mathbf{P}^1) = 2\nu_6(Y').$$

Cancelling two in $\mathbf{Z}$ proves the stabilization assertion. $\square$

Proposition 4.12 (The cubic packet). *For every smooth cubic threefold X ,*

$$\nu_6(X) = 2. \quad (4.9)$$

Proof. Fix the positive generator ℓ of the numerical curve lattice of X , normalized by $P \cdot \ell = 1$, and write $q = Q^\ell$. Thus the coefficient ring is $\mathbf{C}[[q]]$, with the calculation taking place after extension to $\mathcal{K}_X = \overline{\text{Frac}} \mathbf{C}[[q]]$ as in the convention at the start of this section. Cai's small even horizontal system, written in the ordered classical basis $1, P, P^2, P^3$, is

$$\begin{aligned} z^2 \partial_z S &= (K_X + zG_X)S, \\ K_X &= 2 \begin{pmatrix} 0 & 6q & 0 & 36q^2 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \\ G_X &= \frac{1}{2} \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix}. \end{aligned} \quad (4.9a)$$

Here S is the column of coordinates in that basis. This is the original- z -disc frame: any later occurrence of $q^{1/2}$ is an algebraic Novikov-coefficient extension inside $\mathcal{K}_X$, not a ramification of z . Equation (4.9a) is the displayed system in Cai, Section 3, p. 4 [4].

Put $r = (3q)^{1/2} \in \mathcal{K}_X$. The constant change of basis

$$C = \begin{pmatrix} 6r^3 & -6r^3 & 0 & -7r^2 \\ 7r^2 & 7r^2 & -2r^2 & 0 \\ 3r & -3r & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}, \quad \det C = -486r^5, \quad S = C\hat{S} \quad (4.9b)$$

puts the constant term into the form

$$K_0 = C^{-1}K_X C = \begin{pmatrix} 6r & 0 & 0 & 0 \\ 0 & -6r & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad G_0 = C^{-1}G_X C. \quad (4.9c)$$

Partition the coordinates as $1 \mid 1 \mid 2$. There is a unique formal gauge

$$A(z) = I + \sum_{n \geq 1} A_n z^n, \quad A_n \in \text{Mat}_4(\mathbf{C}[r, r^{-1}]), \quad (4.9d)$$

with zero block-diagonal part for every A_n , such that $\hat{S} = A(z)\tilde{S}$ makes every coefficient block diagonal. Indeed, at order n , the block-off-diagonal coefficient is removed by a Sylvester equation $[K_0, A_n] = B_n$ determined by the preceding orders. The three diagonal blocks of K_0 have pairwise disjoint spectra, so this equation has a unique solution on the block-off-diagonal matrices. In particular, the gauge uses only integral powers of the original loop variable z .

Direct multiplication gives the first reduced coefficients:

$$\begin{aligned} J_0 &= \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \\ D_0 &= \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix}, \\ E_0 &= \begin{pmatrix} 0 & -14/(81r^2) \\ -8/81 & 0 \end{pmatrix}, \\ z^2 \partial_z \tilde{S}_+ &= (6r + O(z^2))\tilde{S}_+, \\ z^2 \partial_z \tilde{S}_- &= (-6r + O(z^2))\tilde{S}_-, \\ z^2 \partial_z \begin{pmatrix} \tilde{S}_3 \\ \tilde{S}_4 \end{pmatrix} &= [J_0 + zD_0 + z^2E_0 + O(z^3)] \begin{pmatrix} \tilde{S}_3 \\ \tilde{S}_4 \end{pmatrix}. \end{aligned} \quad (4.9e)$$

This is the integral- z block reduction used by Cai in Section 3, pp. 4–5 [4]; the displayed change of basis and recurrence make the reduction (4.9e) internal to the present calculation.

For the rank-two block, seek a formal solution in the form

$$\tilde{S}_3 = z^\rho(1 + O(z)), \quad \tilde{S}_4 = cz^{\rho+1}(1 + O(z)). \quad (4.9f)$$

The coefficient of $z^{\rho+1}$ in the first coordinate of (4.9e), and that of $z^{\rho+2}$ in the second, give respectively

$$\rho = 2c - \frac{19}{18}, \quad c(\rho + 1) = \frac{19}{18}c - \frac{8}{81}. \quad (4.9g)$$

Eliminating c gives

$$\frac{1}{2} \left(\rho + \frac{19}{18} \right) \left(\rho - \frac{1}{18} \right) = -\frac{8}{81}, \quad \text{hence} \quad \rho^2 + \rho + \frac{5}{36} = 0. \quad (4.9h)$$

This is the coefficient comparison in Cai's displayed rank-two ansatz [4, p. 6].

The quadratic formula applied to (4.9h) gives

$$\rho = -\frac{1}{6}, -\frac{5}{6}. \quad (4.9i)$$

The constant matrix C adjoins only the algebraic Novikov coefficient r , and (4.9d) and its inverse use only integral powers of z . Hence they preserve each exponent class in $\mathbf{Q}/\mathbf{Z}$: multiplication sends $z^\rho \mathcal{K}_X[[z]]$ to itself, and the same holds for the inverse. Thus the two classes in (4.9i) remain the classes in the original- z -disc frame. By (4.0), the repaired framed convention assigns to a rational exponent ρ the eigenvalue $\text{Exp}_V(\rho) = e^{2\pi i \rho}$. Therefore these two classes give

$$\text{Exp}_V\left(-\frac{1}{6}\right) = e^{-\pi i/3}, \quad \text{Exp}_V\left(-\frac{5}{6}\right) = e^{-5\pi i/3} = e^{\pi i/3}. \quad (4.9j)$$

This agrees with Cai's rank-two calculation [4, p. 6].

Put $R = \mathbf{C}[r, r^{-1}]$. The two scalar equations in (4.9e) have the form

$$z^2 \partial_z \tilde{S}_\pm = (\pm 6r + h_\pm(z)) \tilde{S}_\pm, \quad h_\pm(z) \in z^2 R[[z]]. \quad (4.9k)$$

Their normalized formal solutions are

$$\begin{aligned} \tilde{S}_+ &= e^{-6r/z} u_+(z), \quad u_+(z) \in 1 + zR[[z]], \\ \tilde{S}_- &= e^{6r/z} u_-(z), \quad u_-(z) \in 1 + zR[[z]]. \end{aligned} \quad (4.9l)$$

Indeed, after the displayed exponential is removed, $\partial_z u_\pm = z^{-2} h_\pm(z) u_\pm$. The coefficient $z^{-2} h_\pm(z)$ lies in $R[[z]]$, so integration with initial value one gives the asserted units. The exponentials and these units use only integral powers of the original z ; neither adjoins a root of z nor produces a deck factor. Thus each block is unramified and has framed regular-monodromy eigenvalue 1, so neither contributes a primitive sixth root. This agrees with Cai's scalar solutions [4, pp. 5–6].

It remains to count the rank-two solutions. Write its ansatz as $\tilde{S}_3 = z^\rho \sum_{n \geq 0} a_n z^n$ and $\tilde{S}_4 = z^{\rho+1} \sum_{n \geq 0} b_n z^n$, with $a_0 = 1$. At coefficient n , the two equations have the form

$$L_{\rho+n} \begin{pmatrix} a_n \\ b_n \end{pmatrix} = \begin{pmatrix} \alpha_n \\ \beta_n \end{pmatrix}, \quad L_s = \begin{pmatrix} s + 19/18 & -2 \\ 8/81 & s - 1/18 \end{pmatrix}, \quad \det L_s = \left(s + \frac{1}{6}\right) \left(s + \frac{5}{6}\right). \quad (4.9m)$$

Here (α_n, β_n) depends only on coefficients of lower index; for $n = 0$ it is zero, and (4.9h) is exactly the condition that L_ρ have a kernel. The roots in (4.9i) are distinct modulo $\mathbf{Z}$, so $L_{\rho+n}$ is invertible for every $n \geq 1$. For either root, $\ker L_\rho$ is one-dimensional and its first coordinate is nonzero; the normalization $a_0 = 1$ fixes b_0 . The recursion therefore gives one formal solution for each root and no further solution in its exponent class. The two resulting solutions are independent, and exhaust the rank-two block.

Consequently each eigenvalue in (4.9j) has algebraic multiplicity one, while (4.9l) contributes neither primitive sixth root. Hence

$$\nu_6(X) = 2. \quad (4.9n)$$

This is a calculation in the small even connection, so no odd-cohomology block enters the count. Moreover, $N_1(X) = \mathbf{Z}\ell$, and the numerical Novikov ring is exactly $\mathbf{C}[[Q^\ell]] = \mathbf{C}[[q]]$; thus the numerical-Novikov convention does not alter the calculation.

Cai's Section 3 supplies the original small-even matrix on p. 4 and serves as a line-by-line comparison for the remaining calculation: the constant and integral- z reductions appear on pp. 4–5, the rank-two ansatz on p. 6, and the scalar solutions on pp. 5–6. Starting from (4.9a), the change of basis, gauge construction, recursion, and framed count are proved through (4.9n); no assertion about Cai's gauge is used as an input. $\square$

Proof of Theorem 1.1. The projective-bundle formula and Proposition 4.12 give

$$\nu_6(X \times \mathbf{P}^1) = 2\nu_6(X) > 0. \quad (4.10)$$

Suppose that $X \times \mathbf{P}^1$ were rational. By Theorem 4.11,

$$\nu_6(X \times \mathbf{P}^1) = \nu_6(\mathbf{P}^4) = 0,$$

where the last equality follows from (4.2) over a point. This contradicts (4.10). $\square$

4.1 A genus-eight Fano consequence

Kuznetsov associates to every smooth prime Fano threefold V of genus eight a smooth Pfaffian cubic threefold X , together with rank-two vector bundles U on V and E^* on X , whose projectivizations are related by a flop [15, Theorems 2.17–2.18].

Corollary 4.13 (One stabilization of V_{14}). *For every smooth prime Fano threefold V of genus eight,*

$$\nu_6(V) = 2, \quad V \times \mathbf{P}^1 \text{ is irrational.} \quad (4.11)$$

Proof. The flop gives a birational map between the smooth projective fourfolds $\mathbf{P}_V(U)$ and $\mathbf{P}_X(E^*)$. By Theorem 4.11, Proposition 4.7, and Proposition 4.12,

$$2\nu_6(V) = \nu_6(\mathbf{P}_V(U)) = \nu_6(\mathbf{P}_X(E^*)) = 2\nu_6(X) = 4.$$

Hence $\nu_6(V) = 2$. Formula (4.2) now gives $\nu_6(V \times \mathbf{P}^1) = 4$, whereas every rational smooth fourfold has $\nu_6 = 0$ by Theorem 4.11. $\square$

The argument is sharp for this invariant. In a factorization of the fivefold $X \times \mathbf{P}^2$, a center may have dimension three and may itself carry the cubic packet. The next stabilization therefore requires information about the packet's Stokes or Rees placement, not only its multiplicity.

5 The separation theorem

Each separation statement of the introduction has the same two-line proof: a source supplies universal CH_0 -triviality of the threefold, the projective bundle formula for Chow groups carries it to the product, and Theorem 1.1 supplies irrationality.

Proof of Corollary 1.2. Voisin produces a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds along which the primitive minimal class $\Theta^4/4!$ is algebraic, hence along which CH_0 is universally trivial [20, Theorem 4.5 and Lemma 4.6]. $\square$

Proof of Corollary 1.3. The Fermat equation is a sum of five forms in separated single variables, so [6] applies. $\square$

Proof of Corollary 1.4. The cubics of [22, Theorem 3.3] carry unirational parametrizations of degrees two and three, and [22, Corollary 3.5] keeps both degrees on $X \times \mathbf{P}^m$. Unirational parametrizations of coprime degrees give a decomposition of the diagonal, hence universal CH_0 -triviality [22, Section 1]. $\square$

Proof of Theorem 1.5. Corollary 3.6 gives universal CH_0 -triviality of X_b . The projective bundle formula for Chow groups gives the same property for $X_b \times \mathbf{P}^1$. Theorem 1.1 proves that this fourfold is irrational. Non-isotriviality follows from the nonconstant intermediate-Jacobian period map on the A_5 -component: Hartlieb proves that the closure of its image is a one-dimensional special subvariety [9, Proposition 5.7]. $\square$

The algebraic class $\Theta^4/4!$ removes the decomposition-of-diagonal obstruction, while the primitive-sixth framed-monodromy multiplicity survives one stabilization and obstructs a rational factorization. Both conclusions hold on the A_5 -family. They also hold on the coprime-degree family of [22], where the first one comes from a decomposition of the diagonal rather than from an algebraic minimal class.

The two detectors are independent already on smooth projective threefolds. If $U(Y)$ is the indicator of universal CH_0 -triviality, then

$$(U(Y), \nu_6(Y)) = \begin{cases} (1, 0), & Y = \mathbf{P}^3, \\ (1, 2), & Y \text{ is a smooth member of the family above,} \\ (0, 2), & Y \text{ is a very general cubic threefold.} \end{cases}$$

Voisin’s criterion [20, Corollary 4.4] and the evenness theorem [7, Theorem 1.3 and Corollary 1.4] give the last line. Thus neither universal CH_0 -triviality nor ν_6 determines the other.

For $X \times \mathbf{P}^m$ with $m \geq 2$, weak-factorization centers may have dimension at least three and can carry primitive-sixth quantum packets. The low-dimensional vanishing in Proposition 4.10 no longer proves noncancellation. Likewise, the fibrewise minimal-class argument gives no relative universal cycle over the pencil. These are separate problems and are not implicit in the theorem above.

Vanishing of the decomposition-of-the-diagonal obstruction therefore does not predict rationality after one explicit stabilization.

Numbered statements are cited from the exact versions listed here. Where both a journal and a preprint version appear, the numbering is that of the preprint.

References

- [1] D. Abramovich, K. Karu, K. Matsuki, and J. Włodarczyk, *Torification and factorization of birational maps*, J. Amer. Math. Soc. 15 (2002), 531–572.
- [2] J. D. Achter, S. Casalaina-Martin, and J. Wise, *Images of abelian schemes*, preprint, arXiv:2312.13262v2 (2024).
- [3] T. Beckmann and O. de Gaay Fortman, *Integral Fourier transforms and the integral Hodge conjecture for one-cycles on abelian varieties*, Compos. Math. 159 (2023), 1188–1213.
- [4] J. Cai, *The cubic threefold is symplectically irrational*, arXiv:2608.01577v1 (2026).
- [5] C. H. Clemens and P. A. Griffiths, *The intermediate Jacobian of the cubic threefold*, Ann. of Math. (2) 95 (1972), 281–356.
- [6] J.-L. Colliot-Thélène, *CH_0 -trivialité universelle d’hypersurfaces cubiques presque diagonales*, Algebr. Geom. 4 (2017), 597–602; arXiv:1607.05673v3.
- [7] P. Engel, O. de Gaay Fortman, and S. Schreieder, *Matroids and the integral Hodge conjecture for abelian varieties*, arXiv:2507.15704v3 (2026).
- [8] N. Grieve, *Reduced norms and the Riemann–Roch theorem for abelian varieties*, New York J. Math. 23 (2017), 1087–1110; arXiv:1603.06425v2.

- [9] M. Hartlieb, *Special subvarieties in the locus of intermediate Jacobians of cubic threefolds*, Math. Z. 310 (2025), article 52; arXiv:2304.03214v2.
- [10] K. Jansen, K. Lux, R. Parker, and R. Wilson, *An atlas of Brauer characters*, London Mathematical Society Monographs, New Series 11, Oxford University Press, 1995; see also the A_5 representation table, version 3.004, especially the matrix rows with characteristic/ring/dimension/ID 2/ GF(4)/2/ a, b , 2/ GF(2)/4/ a, b , and 3/ GF(3)/4, at <https://brauer.maths.qmul.ac.uk/Atlas/v3/alt/A5/>.
- [11] H. Iritani, *Quantum cohomology of blowups*, arXiv:2307.13555v3 (2025).
- [12] H. Iritani, *Notes on the decomposition theorem for blowups*, arXiv:2604.10028v2 (2026).
- [13] H. Iritani and Y. Koto, *Quantum cohomology of projective bundles*, arXiv:2307.03696v4 (2026).
- [14] L. Katzarkov, M. Kontsevich, T. Pantev, and T. Y. Yu, *Birational invariants from Hodge structures and quantum multiplication*, arXiv:2508.05105v2 (2026).
- [15] A. Kuznetsov, *Derived categories of cubic and V_{14} threefolds*, Proc. Steklov Inst. Math. 246 (2004), 171–194; arXiv:math/0303037v1.
- [16] J. S. Milne, *Lefschetz classes on abelian varieties*, Duke Math. J. 96 (1999), 639–675.
- [17] B. Moonen and A. Polishchuk, *Divided powers in Chow rings and integral Fourier transforms*, Adv. Math. 224 (2010), 2216–2236.
- [18] X. Roulleau, *Genus 2 curve configurations on Fano surfaces*, Comment. Math. Univ. St. Pauli 59 (2010), 51–64; arXiv:1002.4467v1.
- [19] C. Sabbah, *Introduction to Stokes Structures*, Lecture Notes in Mathematics 2060, Springer, 2013; arXiv:0912.2762.
- [20] C. Voisin, *On the universal CH_0 group of cubic hypersurfaces*, J. Eur. Math. Soc. 19 (2017), 1619–1653; arXiv:1407.7261v2.
- [21] J. Yu, *Tropicalizing the positive semidefinite cone*, Proc. Amer. Math. Soc. 143 (2015), 1891–1895.
- [22] S. Yang, X. Yu, and Z. Zhu, *Nonrational varieties with unirational parametrizations of coprime degrees*, arXiv:2508.03623v2 (2025).